Technology Stack

Date	13 November 2026
Team ID	0157CS241131
Project Name	AI Travel Itinerary Planner (Trip Genius AI)
Maximum Marks	2 Marks

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React 19, TypeScript 5.8, Vite 6	Reusable components, type checking, fast dev server and a quick build
2	Backend / Server-Side	Python, FastAPI, Pydantic, Uvicorn + Gunicorn	Fast API framework with built-in validation and Swagger docs, and easy to use with the Groq SDK
3	Database / Data Storage	Supabase PostgreSQL (JSONB + RLS)	Tables for users and trips, JSONB for the itinerary content and row level security
4	Cloud / Hosting / Deployment	Vercel (frontend), Render (backend), Supabase (managed)	Free plans available, deploys from GitHub, HTTPS by default
5	Version Control & CI/CD	Git, GitHub; auto-deploy via Vercel and Render	Keeps code history and redeploys automatically on every push
6	Third-Party APIs / Other Tools	Groq API (openai/gpt-oss-120b), Supabase Auth, jsPDF, pytest, httpx	AI itinerary generation, authentication, PDF export, automated testing, HTTP calls

Dependencies

Layer	Packages
Frontend	react ^19.1, react-dom ^19.1, @supabase/supabase-js ^2.118, jspdf ^4.2, vite ^6.3, typescript ~5.8
Backend	fastapi, uvicorn[standard], gunicorn, pydantic-settings, python-dotenv, httpx, groq, pytest